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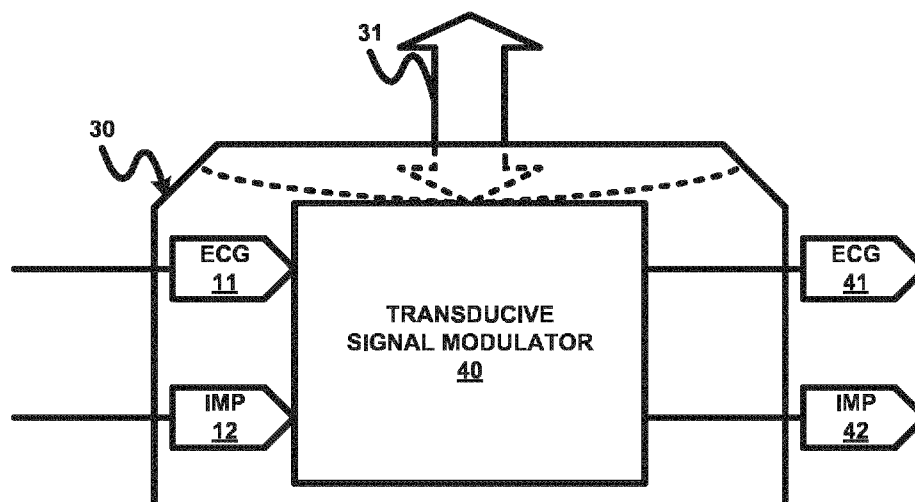
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(54) Title: TRANSDUCIVE SIGNAL MODULATOR FOR SIMULATED CARDIOPULMONARY RESUSCITATION



**FIG. 2A**

(57) Abstract: A cardiopulmonary resuscitation simulation system employing a CPR manikin (30) and a transductive signal modulator (40). In operation, the transductive signal modulator (40) derives a corrupted simulated electrocardiogram signal from a signal modulation of a clean simulated electrocardiogram signal electromechanically responsive to an oscillating compression depth of the CPR manikin (30) during a simulated application of the cardiopulmonary resuscitation to the CPR manikin (30), and further derives the corrupted simulated thoracic impedance signal from a signal modulation of a clean simulated thoracic impedance signal electromechanically responsive to an oscillating deformation of the CPR manikin (30) during the simulated application of the cardiopulmonary resuscitation to the CPR manikin (30).

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## TRANSDUCIVE SIGNAL MODULATOR FOR SIMULATED CARDIOPULMONARY RESUSCITATION

### FIELD OF THE INVENTION

The present disclosure generally relates to a simulated cardiopulmonary resuscitation (CPR) applied to a CPR manikin. The present disclosure specifically relates to a modulation of signals when chest compressions are performed during the simulated CPR.

### BACKGROUND OF THE INVENTION

US 2023/255537 A1 discloses simulating the change of chest impedance with passive components and electrocardiogram contamination. An electromechanical system for generating a CPR-corrupted ECG signal is provided therein.

Cardiopulmonary resuscitation (CPR) is lifesaving intervention that is provided when a patient is in cardiac arrest (e.g., with no or ineffective mechanical activity of the patient's heart) and primarily consists of chest compressions often combined with ventilation and the administration of medication. A quality of CPR is dependent on various factors including maintaining an adequate compression rate and compression depth and avoiding excessive ventilation. As such, training responders via a simulated application of CPR to a CPR manikin is well known in the art of the present disclosure.

During an application of CPR to the patient, defibrillator can be attached to a patient 10 by an electrode pairing to thereby conditionally deliver defibrillating shock(s) to the patient. More specifically, the defibrillator can deliver a high-voltage impulse to the heart of patient in order to restore normal rhythm and contractile function for the heart of patient. As such, a testing of the defibrillator when not in use with CPR is mandated to ensure an operational efficiency of the defibrillator when in use with CPR.

For example, FIG. 1A illustrates a signal generator 10 and an automatic external defibrillator (AED) 20. For purposes of testing the AED 20 when not is use with CPR as known in the art of the present disclosure, signal generator 10 is operated to generate a clean electrocardiogram (ECG) signal 11 (e.g., clean ECG signal 11a of FIG. 1B) and a clean thoracic impedance signal 12 (e.g., clean thoracic impedance signal 12a as shown in FIG. 1C), both excluding any artifacts associated with CPR being applied to a patient (i.e. CPR artifacts).

As known in the art of the present disclosure, AED 20 analyzes clean ECG signal 11 for any shockable rhythm(s) (e.g., ventricular fibrillation or ventricular tachycardia) and signal generator 10 temporally generates clean ECG signal 11 with and without shockable rhythms to ascertain how effective AED 20 is in detecting the shockable rhythms within clean ECG signal 11.

Also as known in the art of the present disclosure, AED 20 further compares the clean thoracic impedance signal 12 to a threshold delineating a probable successful defibrillation or an improbable successful defibrillation based on an impedance between an electrode pairing attached to the patient, and signal generator 20 temporally generates clean thoracic impedance signal 12 at various flat levels to ascertain how effective AED 20 is in comparing the clean thoracic impedance signal 12 to the threshold.

## SUMMARY OF THE INVENTION

The present disclosure is directed to an operational demonstration and/or testing of a defibrillator (e.g., an AED, a semi-automatic defibrillator and a manual defibrillator) during a simulated application of CPR to a CPR manikin. The invention is defined by the independent claims. The dependent claims define advantageous embodiments.

The present disclosure can be embodied as (1) a transductive signal modulator for generating a corrupted simulated electrocardiogram signal and a corrupted simulated thoracic impedance signal during a simulated application of a CPR to a CPR manikin, (2) a CPR simulation system for operationally demonstrating and/or testing a defibrillator based on a generation of a corrupted simulated electrocardiogram signal and a corrupted simulated thoracic impedance signal during a simulated application of a CPR to a CPR manikin, and (3) a CPR simulation method for operationally demonstrating and/or testing a defibrillator based on a generation of a corrupted simulated electrocardiogram signal and a corrupted simulated thoracic impedance signal during a simulated application of a CPR to a CPR manikin.

Various exemplary embodiments of a transductive signal modulator of the present disclosure employ a depth transducer and a deformation transducer.

The depth transducer is internally disposed within the CPR manikin to electromechanically respond to an oscillating compression depth of the CPR manikin during the simulated application of the cardiopulmonary resuscitation to the CPR manikin. The depth transducer is configured to derive the corrupted simulated electrocardiogram signal from a signal modulation of a clean simulated electrocardiogram signal electromechanically responsive to the oscillating compression depth of the CPR manikin.

The deformation transducer is internally disposed within the CPR manikin to electromechanically respond to an oscillating deformation of the CPR manikin during the simulated application of the cardiopulmonary resuscitation to the CPR manikin. The deformation transducer is configured to derive the corrupted simulated thoracic impedance signal from a signal modulation of a clean simulated thoracic impedance signal electromechanically responsive to the oscillating deformation of the CPR manikin.

Various exemplary embodiments of a CPR simulation system of the present disclosure employs an aforementioned transductive signal modulator of the present disclosure and a CPR manikin.

Various exemplary embodiments of a CPR simulation system of the present disclosure can further employ a signal generator for generating the clean ECG signal and the clean thoracic impedance signal during the simulated application of the cardiopulmonary resuscitation to the CPR manikin, and/or a defibrillator for analyzing the corrupted ECG signal and the corrupted thoracic impedance signal during or subsequent to the simulated application of the cardiopulmonary resuscitation to the CPR manikin.

Various exemplary embodiments of a CPR simulation method of the present disclosure involve (1) an electromechanical responding, by a transductive signal modulator internally affixed within a CPR manikin, of an oscillating compression depth of the CPR manikin during the simulated application of the cardiopulmonary resuscitation to the CPR manikin, (2) a deriving, by the transductive signal modulator, of a corrupted simulated electrocardiogram signal from a signal modulation of the clean simulated electrocardiogram signal electromechanically responsive to the oscillating compression depth of the CPR manikin, (3) an electromechanical responding, by the transductive signal modulator, of an oscillating deformation of the CPR manikin during the simulated application of the cardiopulmonary resuscitation to the CPR manikin, and (4) a deriving, by the transductive signal modulator, of a corrupted simulated thoracic impedance signal from a signal modulation of the clean simulated thoracic impedance signal electromechanically responsive to the oscillating deformation of the CPR manikin.

Various exemplary embodiments of a CPR simulation method of the present disclosure can further involve (5) a generating, by a signal generator, of the clean simulated electrocardiogram signal and the clean simulated thoracic impedance signal during a simulated application of the cardiopulmonary resuscitation to the CPR manikin, and/or (6) operationally demonstrating and/or testing a defibrillator based on the corrupted simulated electrocardiogram signal and the corrupted simulated thoracic impedance signal during or subsequent to the simulated application of the cardiopulmonary resuscitation to the CPR manikin.

The foregoing exemplary embodiments and other embodiments of the present disclosure as well as various structures and advantages of the present disclosure will become further apparent to those having ordinary skill in the art from the following detailed description of various embodiments of the present disclosure read in conjunction with the accompanying drawings. The detailed description and drawings are merely illustrative of the present disclosure rather than limiting, the scope of the present disclosure being defined by the appended claims and equivalents thereof.

## BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will present in detail the following description of exemplary embodiments with reference to the following figures wherein:

FIG. 1A illustrates an exemplary defibrillation demonstration/testing system as known in the art of the present disclosure

FIG. 1B illustrates an exemplary clean ECG signal and an exemplary clean thoracic impedance signal as known in the art of the present disclosure;

FIG. 2A illustrates an exemplary cardiopulmonary resuscitation system in accordance with present disclosure;

FIG. 2B illustrates an exemplary corrupted ECG signal and an exemplary corrupted thoracic impedance signal in accordance with the art of the present disclosure;

FIGS. 3A and 3B illustrate an exemplary depth transducer in accordance with the present disclosure.

FIGS. 4A-4C illustrate exemplary embodiments of the depth transducer of FIGS. 3A and 3B.

FIGS. 5A and 5B illustrate an exemplary deformation transducer in accordance with the present disclosure.

FIGS. 6A-6C illustrate exemplary embodiments of the deformation transducer of FIGS. 5A and 5B.

FIGS. 7A and 7B illustrate exemplary embodiments of a dual transducer in accordance with the present disclosure.

FIG. 7C illustrate exemplary embodiments of a depth sensor and dual signal combiner in accordance with the present disclosure.

## DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present disclosure is directed to electromechanical based devices, systems and methods for simulating real patient behavior when chest compressions are performed during cardiopulmonary resuscitation (CPR) by adding a motion artifact to the thoracic impedance signal and to the ECG signal seen across defibrillator electrode pads.

To facilitate an understanding of the present disclosure, the following description of FIG. 2A-6C teaches devices, systems and methods in accordance with the present disclosure for making and using various embodiments of the present disclosure. From the following description of FIGS. 2A-6C, those having ordinary skill in the art of the present disclosure will appreciate how to apply the present disclosure to make and use additional embodiments of devices, systems and methods in accordance with the present disclosure.

Referring to FIG. 2A, a cardiopulmonary resuscitation simulation system of the present disclosure employs a CPR manikin 30 as known in the art of the present disclosure and a transductive signal modulator 40 in accordance with the present disclosure for generating a corrupted simulated electrocardiogram (ECG) signal 41 (e.g., signal 41a of FIG. 2B) and a corrupted simulated thoracic impedance signal 42 (e.g., signal 42a of FIG. 2B) during a simulated application of a cardiopulmonary resuscitation to CPR manikin 30.

More particularly as shown in FIG. 1A, transductive signal modulator 40 is disposed within CPR manikin 30 to electromechanically respond to an oscillating compression motion 31 of the CPR manikin 30 during a simulated CPR.

Transductive signal modulator 40 derives corrupted simulated electrocardiogram signal 41 from a signal modulation of clean simulated electrocardiogram signal 11 (FIG. 1A) based on an electromechanical response by transductive signal modulator 40 to an oscillating compression depth of CPR manikin 30 during the simulated CPR.

In one exemplary embodiment for deriving corrupted simulated electrocardiogram signal 41, transductive signal modulator 40 employs a deformation transducer 50 as shown in FIGS. 3A and 3B.

Referring to FIGS. 3A and 3B, depth transducer 50 is internally affixed to a base or a chest plate of CPR manikin 30 (not shown) whereby depth transducer 50 electromechanically responds to an oscillating compression depth 33 resulting from an application of beginning compression force 32a to a maximum compression force 32b. From this electromechanical response, depth transducer 50 executes a signal modulation of clean simulated ECG signal 11 into corrupted simulated ECG signal 41.

In practice, depth transducer 50 is modeled as a signal generator or a level shifter with variable gain, whereby the gain is mapped using depth data.  $\Delta V = K(d) d$ . Nonetheless, a good approximation would be the case where the gain is constant, whereby the sinusoidal chest compressions will produce sinusoidal voltage differentials to thereby establish a one-to-one map.

FIGS. 4A-4B illustrate various embodiments of depth transducer 50.

Referring to FIG. 4A, a depth transducer 50a includes a depth sensor 51a for electromechanically responding to the oscillating compression depth 33 and executes a signal modulation of clean simulated ECG signal 11 into corrupted simulated ECG signal 41a (analog) based on the electromechanical response.

In one exemplary embodiment, depth sensor 51a employs a photoresistor.

Referring to FIG. 4B, a depth transducer 50b includes a depth sensor 51b for electromechanically responding to the oscillating compression depth 33 and executes a signal modulation of clean simulated ECG signal 11 into modulated simulated ECG signal 41b (analog) based on the electromechanical response. Depth transducer 50b further includes a signal conditioner 52 for applying a gain to modulated simulated ECG signal 41b to yield corrupted simulated ECG signal 41c.

In one exemplary embodiment, depth sensor 51b employs a photoresistor and signal conditioner 52 includes a potentiometer in series with a divider.

Referring to FIG. 4C, a depth transducer 50c includes a depth sensor 51c for electromechanically responding to the oscillating compression depth 33 and generates a tracking signal 54 based on the electromechanical response. Depth transducer 50c further includes a signal combiner 53 for combining clean simulated ECG signal 11 and tracking signal 54 to yield corrupted ECG signal 41d.

In one exemplary embodiment, depth sensor 51c employs a time-of flight sensor, an ultrasound sensor, a depth camera or a laser, and signal combiner 53 employs a microcontroller and digital-to-analog converter.

Referring to FIGS. 5A and 5B, deformation transducer 60 is internally affixed to a base or a chest plate of CPR manikin 30 (not shown) whereby deformation transducer 60 electromechanically responds to an oscillating deformation 34 resulting from an application of beginning compression force 32a to a maximum compression force 32b. From this electromechanical response, deformation transducer 60 executes a signal modulation of clean simulated thoracic impedance signal 12 into corrupted simulated thoracic impedance signal 42.

In practice, for optimally affixing deformation transducer 60 with CPR manikin 30, a chest plate of CPR manikin 30 can be removed and the deformation of different regions can be closely observed to identify regions with a specified surface strain of (e.g., approximately 0.008) when the CPR manikin undergoes a specified degree of compression (e.g., two (2) inches). On regions of this strain, a strain gauge 51 may be affixed to map deformation to resistance.

Furthermore in practice, a gauge factor relates strain seen to the ratio of change seen in resistance to the nominal resistance  $\frac{\Delta L}{L} K = \frac{\Delta R}{R}$ . This means the change in resistance for the strain gauge is known (e.g., approximately two (2) ohms for a BF120-10aa strain gauge). As such, a base resistance of the strain gauge (e.g., 120 ohms) is fixed whereby two (2) strain gauges can be wired in parallel, whereby the total resistance of the manikin is now  $60 \pm 1 \Omega$ , which is within the typical range for a real patient.

FIGS. 6A-6B illustrate various embodiments of deformation transducer 60.

Referring to FIG. 6A, a deformation transducer 60a includes a strain gauge 61a for electromechanically responding to the oscillating deformation 34 and executes a signal modulation of clean simulated thoracic impedance signal 12 into corrupted simulated thoracic impedance signal 42a (analog) based on the electromechanical response.

In one exemplary embodiment, strain gauge 61a employs a BF120-10a strain gauge and a network of strain gauges 61a can be utilized.

Referring to FIG. 6B, a deformation transducer 60b includes a strain gauge 61b for electromechanically responding to the oscillating deformation 34 and executes a signal modulation of clean simulated thoracic impedance signal 12 into modulated simulated thoracic impedance signal 42b (analog) based on the electromechanical response. Deformation transducer 60b further includes a signal conditioner 62 for applying a gain to modulated simulated thoracic impedance signal 42b to yield corrupted simulated thoracic impedance signal 42c.

In one exemplary embodiment, signal conditioner 52 includes a potentiometer in series with a divider for strain gauge 61b.

Referring to FIG. 6C, a deformation transducer 60c includes a strain gauge 61c for electromechanically responding to the oscillating deformation 34 and modulated a reference signal 64a



into a modulated reference signal 64b based on the electromechanical response. Deformation transducer 60c further includes a signal combiner 53 for combining clean simulated thoracic impedance signal 12 and modulated reference signal 64b to yield corrupted thoracic impedance signal 42d.

In one exemplary embodiment, signal combiner 53 employs a microcontroller and digital-to-analog converter.

FIGS. 7A and 7B illustrate various embodiments of a dual transducer in accordance with the present disclosure. In practice, as would be appreciated by one skilled in the art of the present disclosure, a dual transducer of the present invention implements a combination of a depth transducer and a deformation transducer as described in the present disclosure.

Referring to FIG. 7A, a dual transducer 150a of the present disclosure is electromechanically responsive to an oscillating compression depth (e.g., oscillating compression depth 33 of FIG. 3B) to execute a signal modulation of clean simulated ECG signal 11 into corrupted simulated ECG signal 41e (analog) based on the electromechanical response, and to further execute a signal modulation of clean simulated thoracic impedance signal 12 into corrupted simulated thoracic impedance signal 42e (analog) based on the electromechanical response.

In one exemplary embodiment, dual transducer 150a employs a photoresistor.

Referring to FIG. 7B, a dual transducer 150b is electromechanically responsive to an oscillating compression depth (e.g., oscillating compression depth 33 of FIG. 3B) to execute a signal modulation of clean simulated ECG signal 11 into modulated simulated ECG signal 41f (analog) based on the electromechanical response, and to further execute a signal modulation of clean simulated thoracic impedance signal 12 into modulated simulated thoracic impedance signal 42f (analog) based on the electromechanical response.

A dual signal conditioner 152 applies a gain to modulated simulated ECG signal 41f to yield corrupted simulated ECG signal 41g and a gain to modulated simulated thoracic impedance signal 42f (analog) to yield corrupted simulated ECG signal 41fg.

In one exemplary embodiment, dual transducer 150b employs a photoresistor and dual signal conditioner 152 includes a potentiometer in series with a divider for the ECG signal 41f and a potentiometer in series with a divider for the impedance signal 41f.

Referring to FIG. 7C, depth sensor 51c is electromechanically responsive to the oscillating compression depth 33 (e.g., oscillating compression depth 33 of FIG. 3B) and generates a tracking signal 154 based on the electromechanical response.

A dual signal combiner 153 combines simulated ECG signal 11 and tracking signal 154 to yield corrupted ECG signal 41h, and further combines clean simulated thoracic impedance signal 12 and tracking signal 154 to yield corrupted thoracic impedance signal 42h.

In one exemplary embodiment, depth sensor 51c employs a time-of flight sensor, an ultrasound sensor, a depth camera or a laser, and signal combiner 53 employs a microcontroller and digital-to-analog converter.

In practice, a CPR manikin can be connected to a defibrillator with the defibrillation shock circuitry enabled or disabled. In one exemplary embodiment when the CPR manikin is connected to a defibrillator with the defibrillation shock circuitry enabled, a module as known in the art of the present disclosure or hereinafter conceived can be incorporated in a cardiopulmonary resuscitation simulation system of the present disclosure to protect the system, particularly a transductive signal modulator of the present disclosure, when a defibrillating shock is applied to the CPR manikin by the defibrillator connected to the CPR manikin whereby the system can tolerate the defibrillating shock.

From the description of FIGS. 2A-6C herein, those having ordinary skill in the art will appreciate the numerous benefits of the present disclosure including, but not limited to, a generation and communication of real-time CPR corrupted signals for demonstrating and testing of defibrillators which can analyze CPR corrupted data, particularly as related to inducing an artifact a defibrillator can recognize as a motion artifact generated by CPR.

The present disclosure has been described with reference to the preferred embodiments. Modifications and alterations may occur to others upon reading and understanding the preceding detailed description. It is intended that the invention be construed as including all such modifications and alterations insofar as they come within the scope of the appended claims or the equivalents thereof.

Further, as one having ordinary skill in the art shall appreciate in view of the teachings provided herein, features, elements, components, etc. disclosed and described in the present disclosure/specification and/or depicted in the appended Figures and/or recited in the claims can be implemented in various combinations of hardware and software, and provide functions which may be combined in a single element or multiple elements. For example, the functions of the various features, elements, components, etc. shown/illustrated/depicted in the Figures and/or recited in the claims can be provided through the use of dedicated hardware as well as hardware capable of executing software in association with appropriate software. When provided by a processor, the functions can be provided by a single dedicated processor, by a single shared processor, or by a plurality of individual processors, some of which can be shared and/or multiplexed. Moreover, explicit use of the term “processor” or “controller” should not be construed to refer exclusively to hardware capable of executing software, and can implicitly include, without limitation, digital signal processor (“DSP”) hardware, memory (e.g., read only memory (“ROM”) for storing software, random access memory (“RAM”), non-volatile storage, etc.) and virtually any means and/or machine (including hardware, software, firmware, combinations thereof, etc.) which is capable of (and/or configurable) to perform and/or control a process.

Moreover, all statements herein reciting principles, aspects, and exemplary embodiments of the present disclosure, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future (e.g., any elements developed that can perform the same or substantially similar functionality, regardless of structure). Thus, for example, it will be appreciated by one having ordinary skill in the art in view of the teachings provided

herein that any block diagrams presented herein can represent conceptual views of illustrative system components and/or circuitry embodying the principles of the invention. Similarly, one having ordinary skill in the art should appreciate in view of the teachings provided herein that any flow charts, flow diagrams and the like can represent various processes which can be substantially represented in computer readable storage media and so executed by a computer, processor or other device with processing capabilities, whether or not such computer or processor is explicitly shown.

Having described preferred and exemplary embodiments of the present disclosure, which embodiments are intended to be illustrative and not limiting, it is noted that modifications and variations can be made by persons having ordinary skill in the art in view of the teachings provided herein, including the appended Figures and claims. It is therefore to be understood that changes can be made in/to the preferred and exemplary embodiments of the present disclosure which are within the scope of the present disclosure and exemplary embodiments disclosed, described and taught herein.

Moreover, it is contemplated that corresponding and/or related systems incorporating and/or implementing the device or such as may be used/implemented in a device in accordance with the present disclosure are also contemplated and considered to be within the scope of the present disclosure. Further, corresponding and/or related method for manufacturing and/or using a device and/or system in accordance with the present disclosure are also contemplated and considered to be within the scope of the present disclosure. In the claims, any reference signs placed between parentheses shall not be construed as limiting the claim. The word "comprising" does not exclude the presence of elements or steps other than those listed in a claim. The word "a" or "an" preceding an element does not exclude the presence of a plurality of such elements. The invention may be implemented by means of hardware comprising several distinct elements, and/or by means of a suitably programmed processor. The computer program product may be software available for download from a server, e.g., via the internet. Alternatively, the computer program product may be a suitable (non-transitory) computer-readable medium on which the instructions are stored, such as an optical storage medium or a solid-state medium, which may or may not be supplied together with or as part of other hardware. In the device claim enumerating several means, several of these means may be embodied by one and the same item of hardware. Measures recited in mutually different dependent claims may advantageously be used in combination.

## CLAIMS:

1. A transductive signal modulator (40) for generating a corrupted simulated electrocardiogram signal and a corrupted simulated thoracic impedance signal during a simulated application of a cardiopulmonary resuscitation to a CPR manikin (30), the transductive signal modulator (40) comprising:
- 5 a depth transducer (50) internally disposed within the CPR manikin (30) to electromechanically respond to an oscillating compression depth of the CPR manikin (30) during the simulated application of the cardiopulmonary resuscitation to the CPR manikin (30),
- wherein the depth transducer (50) is configured to derive the corrupted simulated electrocardiogram signal from a signal modulation of a clean simulated electrocardiogram
- 10 signal electromechanically responsive to the oscillating compression depth of the CPR manikin (30); and
- a deformation transducer (60) internally disposed within the CPR manikin (30) to electromechanically respond to an oscillating deformation of the CPR manikin (30) during the simulated application of the cardiopulmonary resuscitation to the CPR manikin (30),
- wherein the deformation transducer (60) is configured to derive the corrupted
- 15 simulated thoracic impedance signal from a signal modulation of a clean simulated thoracic impedance signal electromechanically responsive to the oscillating deformation of the CPR manikin (30).
2. The transductive signal modulator (40) of claim 1,
- wherein the depth transducer (50) includes a depth sensor (51) internally affixed within
- 20 the CPR manikin (30) to electromechanically respond to the oscillating compression depth of the CPR manikin (30); and
- wherein the depth sensor (51) is configured to modulate the clean simulated electrocardiogram signal into the corrupted simulated electrocardiogram signal as a function of the oscillating compression depth of the CPR manikin (30).
- 25
3. The transductive signal modulator (40) of claim 1,
- wherein the depth transducer (50) includes a depth sensor (51) internally affixed within the CPR manikin (30) to electromechanically respond to the oscillating compression depth of the CPR manikin (30).
- 30

4. The transductive signal modulator (40) of claim 3,  
wherein the depth sensor (51) is configured to modulate the clean simulated  
electrocardiogram signal into the modulated simulated electrocardiogram signal as a function of the  
oscillating compression depth of the CPR manikin (30); and

5 wherein the depth transducer (50) further includes a signal conditioner (52) configured to  
condition the modulated simulated electrocardiogram signal into the corrupted simulated  
electrocardiogram signal.

5. The transductive signal modulator (40) of claim 3,

10 wherein the depth sensor (51) is configured to generate a tracking signal as a function of  
the oscillating compression depth of the CPR manikin (30); and

wherein the depth transducer (50) includes a signal combiner (53) configured to combine  
the clean simulated electrocardiogram signal and the tracking signal into the corrupted simulated  
electrocardiogram signal.

15 6. The transductive signal modulator (40) of claim 1,

wherein the deformation transducer (60) includes a strain gauge (61) internally affixed  
within the CPR manikin (30) to electromechanically respond to the oscillating deformation of the CPR  
manikin (30); and

20 wherein the strain gauge (61) is configured to modulate the clean simulated thoracic  
impedance signal into the corrupted simulated thoracic impedance signal as a function of the oscillating  
deformation of the CPR manikin (30).

7. The transductive signal modulator (40) of claim 1,

25 wherein the deformation transducer (60) includes a strain gauge (61) internally affixed  
within the CPR manikin (30) to electromechanically respond to the oscillating deformation of the CPR  
manikin (30);

wherein the strain gauge (61) is configured to modulate the clean simulated thoracic  
impedance signal into a modulated simulated thoracic impedance signal as a function of the oscillating  
30 deformation of the CPR manikin (30); and

wherein the deformation transducer (60) further includes a signal conditioner (62)  
configured to condition the modulated simulated thoracic impedance signal into the corrupted simulated  
thoracic impedance signal.

8. The transductive signal modulator (40) of claim 1,  
wherein the deformation transducer (60) includes a strain gauge (61) internally affixed  
within the CPR manikin (30) electromechanically respond to the oscillating deformation of the CPR  
manikin (30);

5 wherein the strain gauge (61) is configured to modulate a reference deformation signal  
into a modulated deformation signal as a function of the oscillating deformation of the CPR manikin (30);  
and

wherein the deformation transducer (60) further includes a signal combiner (63)  
configured to combine the clean simulated thoracic impedance signal and the modulated deformation  
10 signal into the corrupted simulated thoracic impedance signal.

9. A cardiopulmonary resuscitation simulation system, comprising:  
a CPR manikin (30); and  
a transductive signal modulator (40) as claimed in any of the preceding claims.

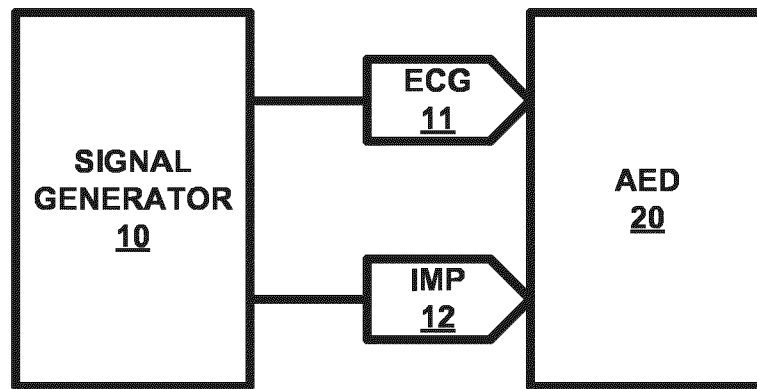
10. A cardiopulmonary resuscitation simulation method, comprising:  
electromechanically responding, by a transductive signal modulator (40) internally affixed  
with a CPR manikin (30), an oscillating compression depth of the CPR manikin (30) during the simulated  
application of the cardiopulmonary resuscitation to the CPR manikin (30),

20 deriving, by the transductive signal modulator (40), a corrupted simulated  
electrocardiogram signal from a signal modulation of the clean simulated electrocardiogram signal  
electromechanically responsive to the oscillating compression depth of the CPR manikin (30);

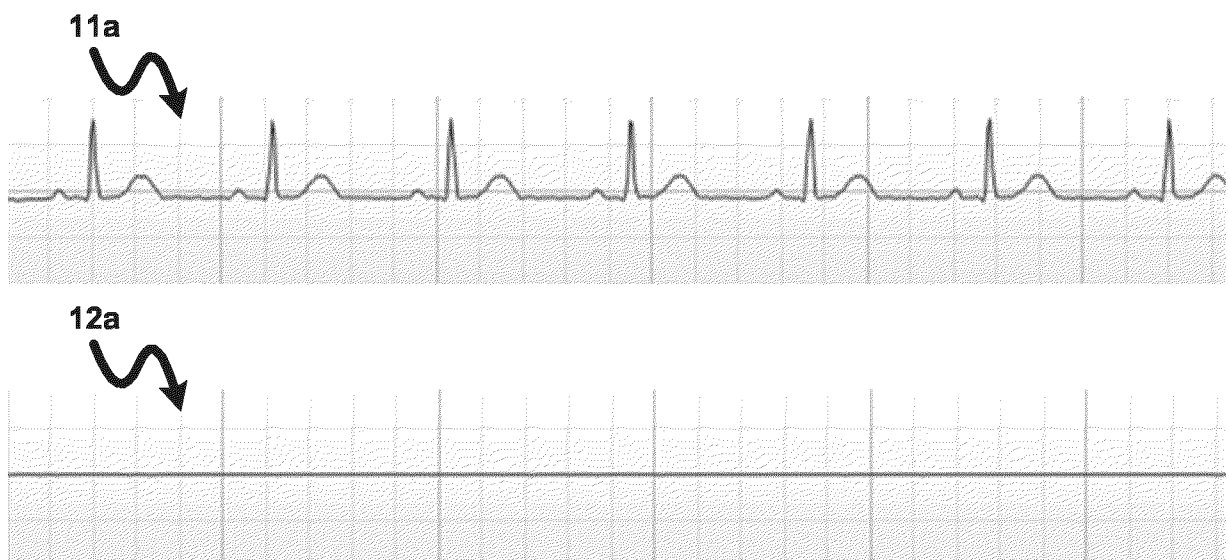
electromechanically responding, by the transductive signal modulator (40), an oscillating  
deformation of the CPR manikin (30) during the simulated application of the cardiopulmonary  
25 resuscitation to the CPR manikin (30); and

deriving, by the transductive signal modulator (40), a corrupted simulated thoracic  
impedance signal from a signal modulation of the clean simulated thoracic impedance signal  
electromechanically responsive to the oscillating deformation of the CPR manikin (30).

30 11. A computer program product for enabling a transductive signal modulator (40) to generate  
a corrupted simulated electrocardiogram signal and a corrupted simulated thoracic impedance signal for a  
cardiopulmonary resuscitation simulation method as claimed in claim 10.



**FIG. 1A**  
**(PRIOR ART)**



**FIG. 1B**  
**(PRIOR ART)**

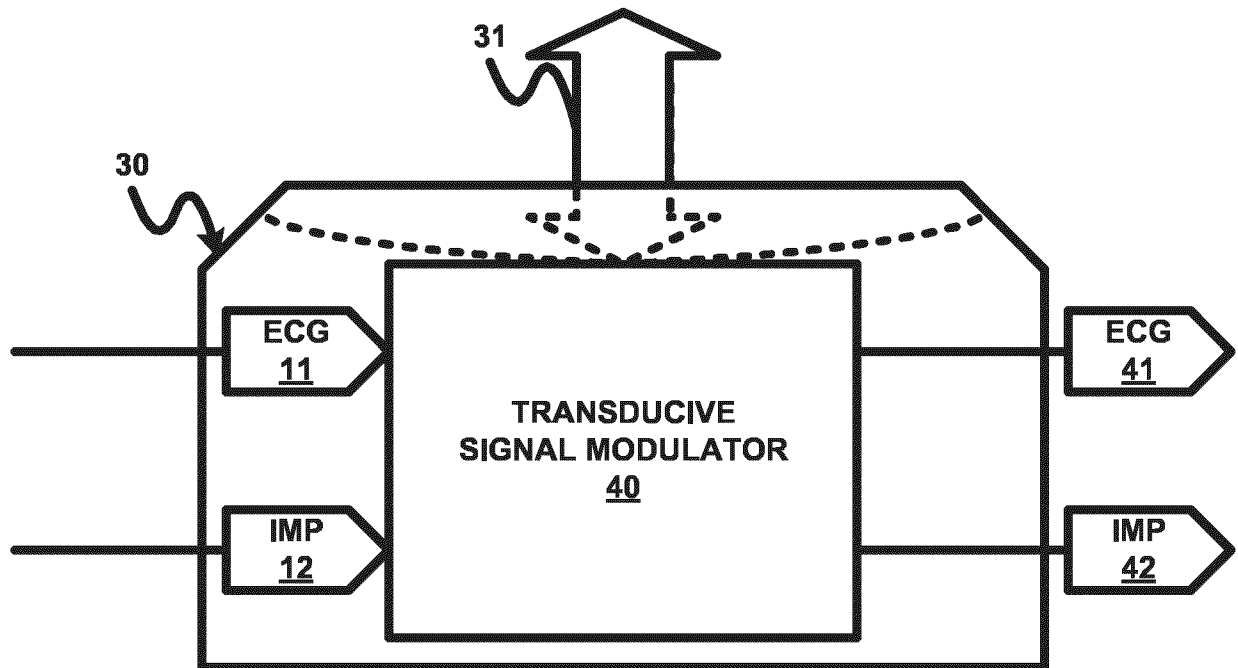


FIG. 2A

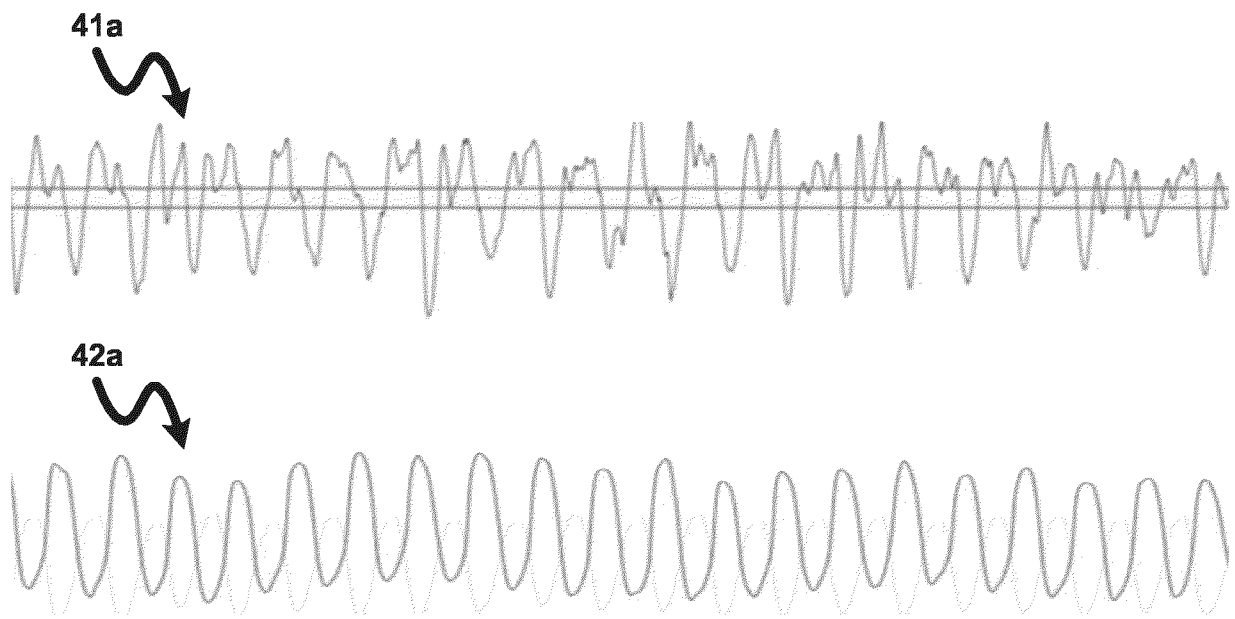


FIG. 2B



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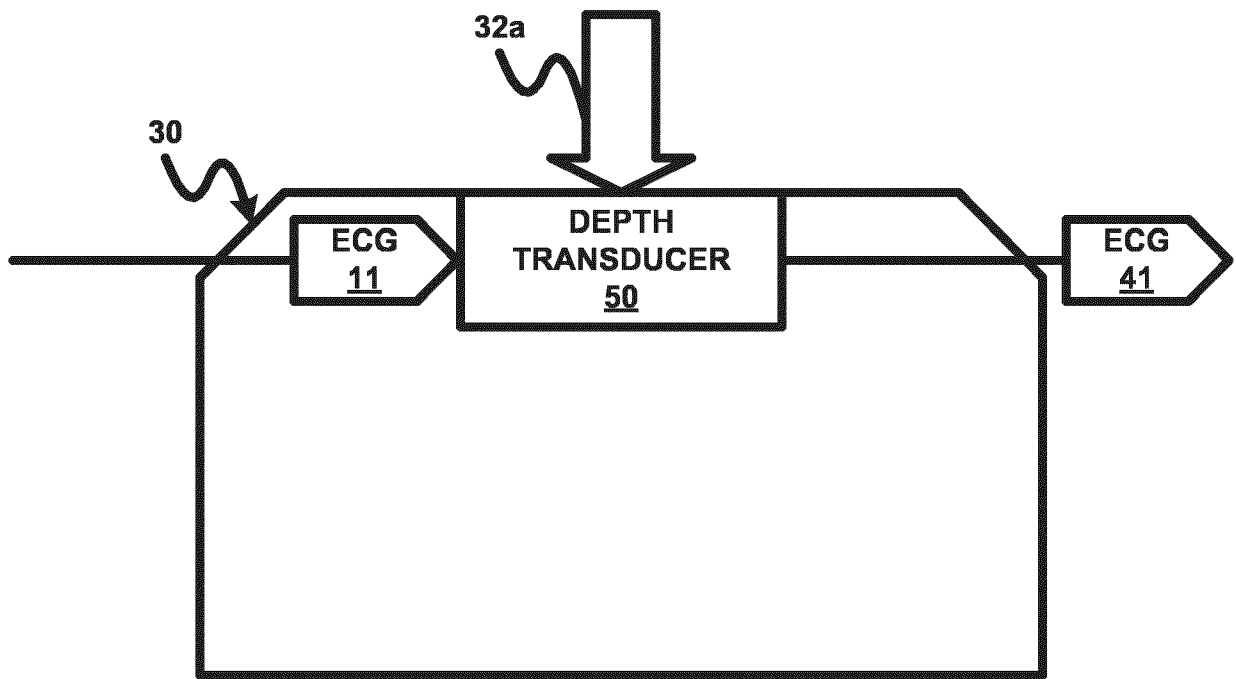


FIG. 3A

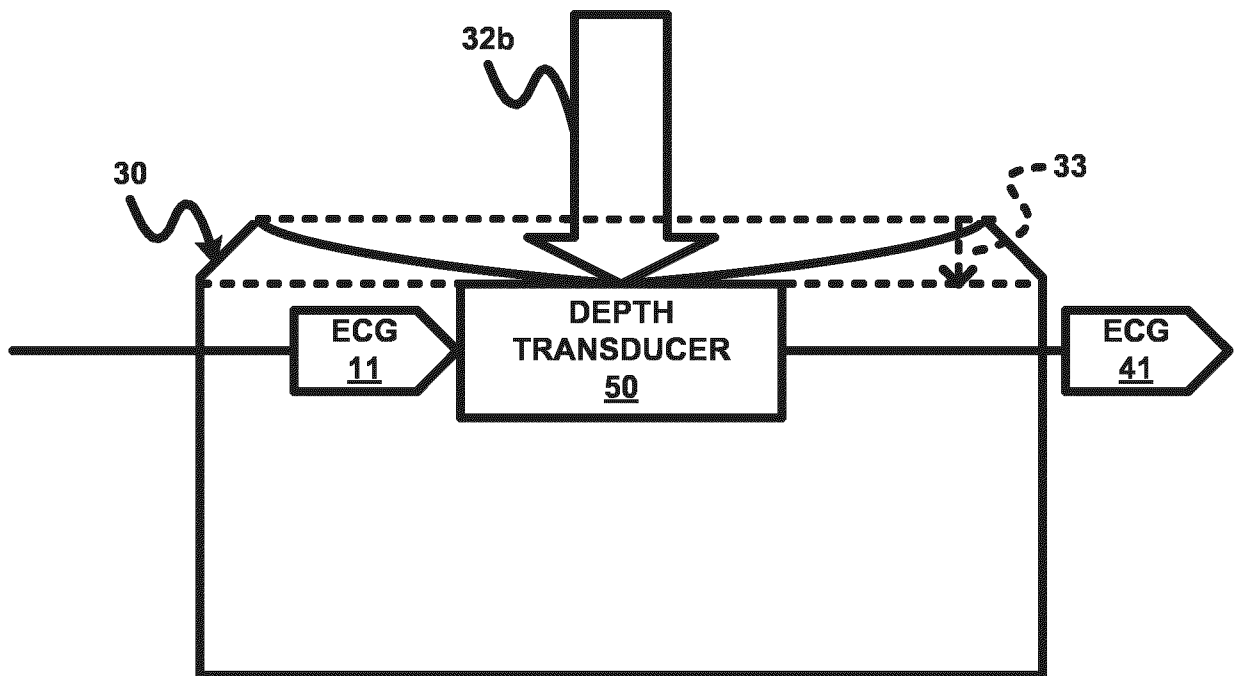


FIG. 3B

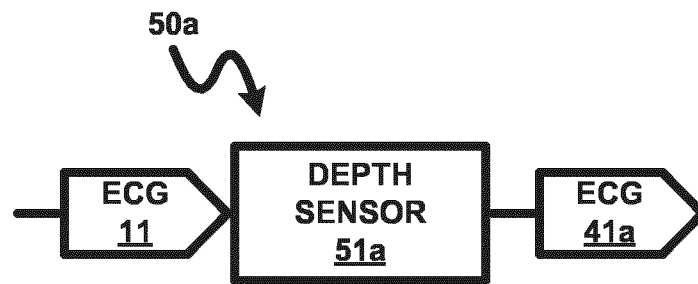


FIG. 4A

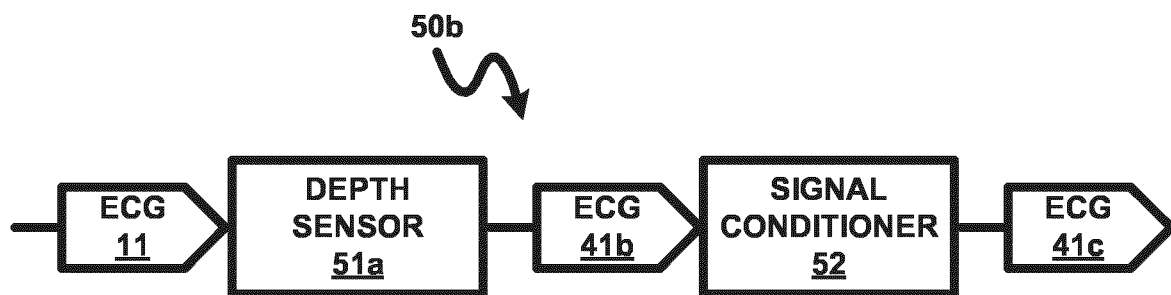


FIG. 4B

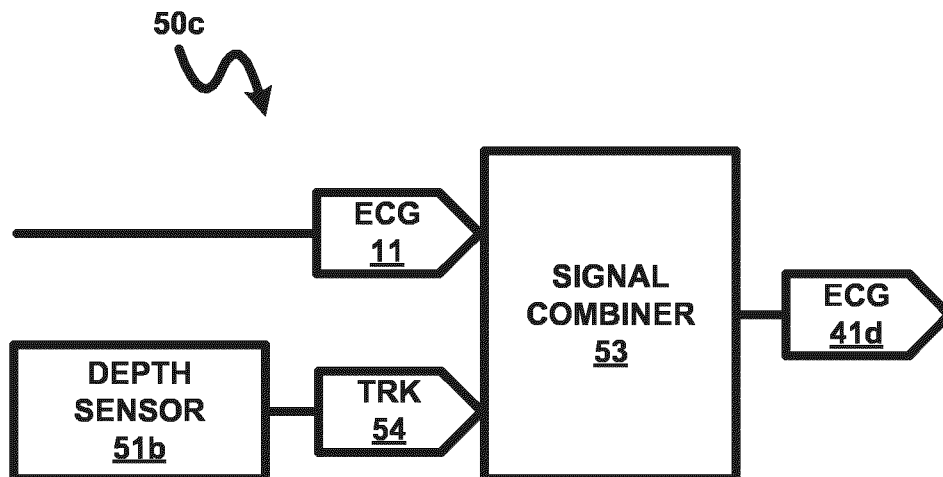


FIG. 4C

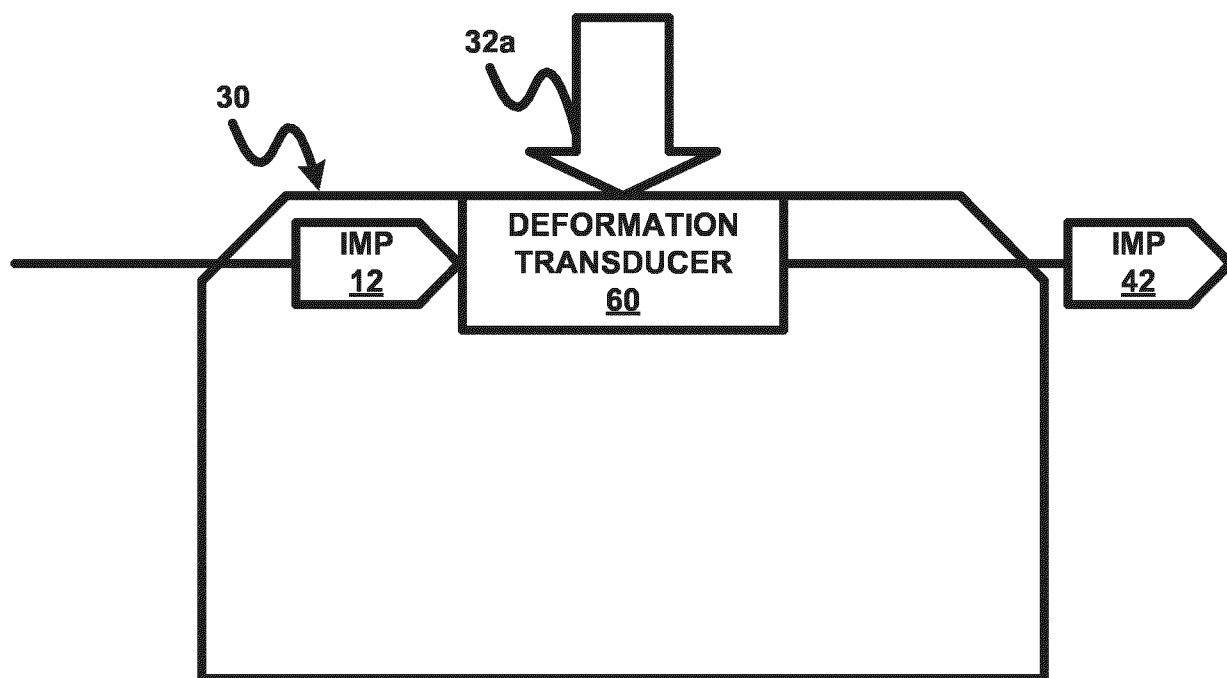


FIG. 5A

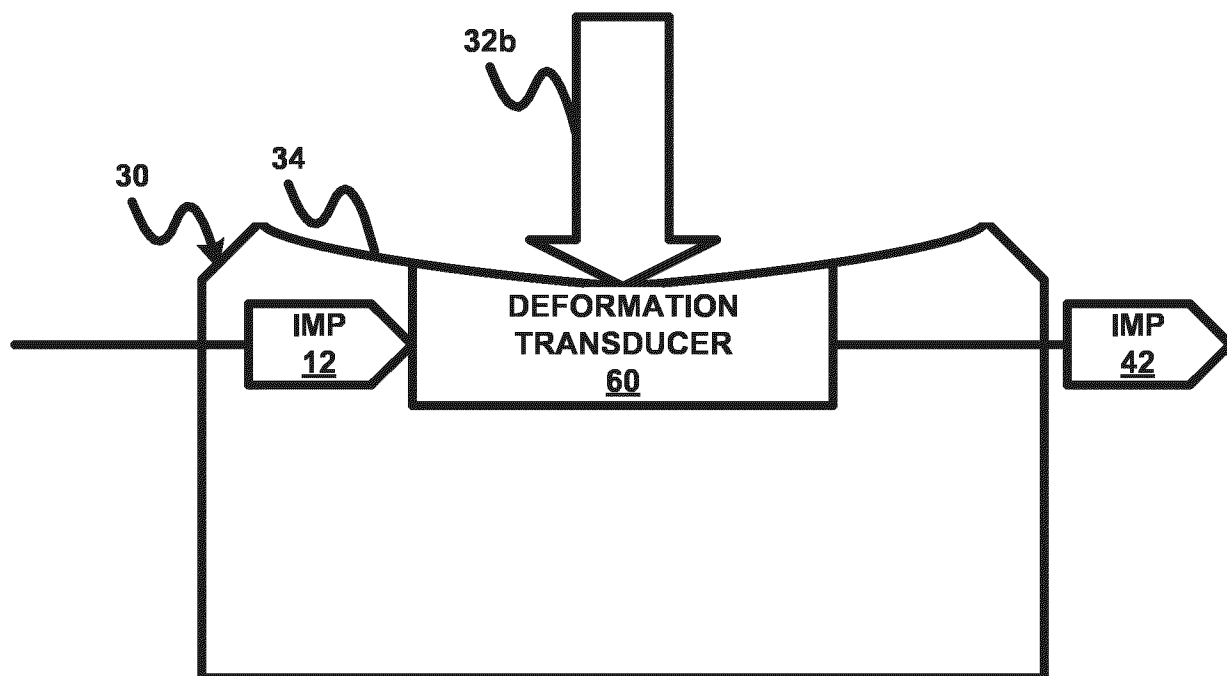


FIG. 5B

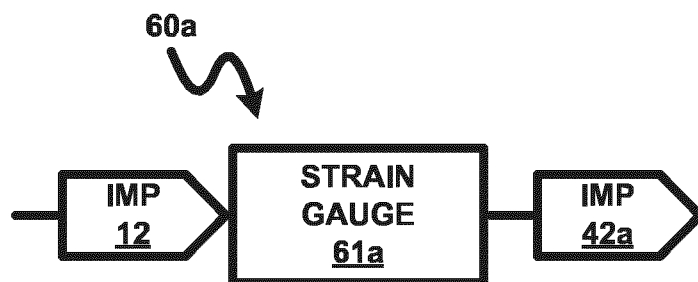


FIG. 6A

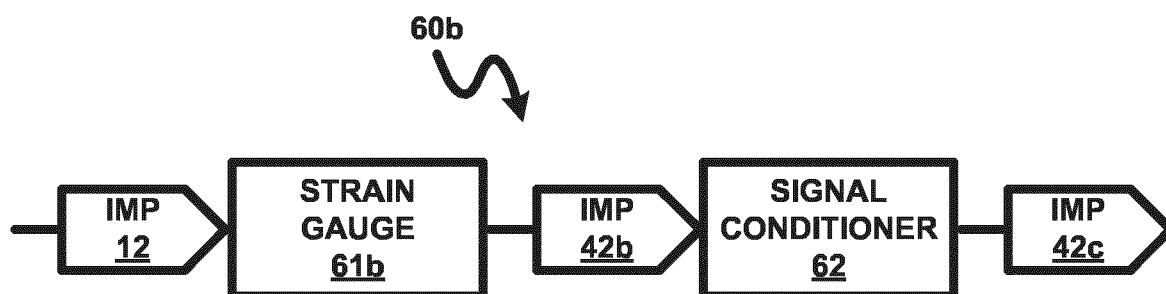


FIG. 6B

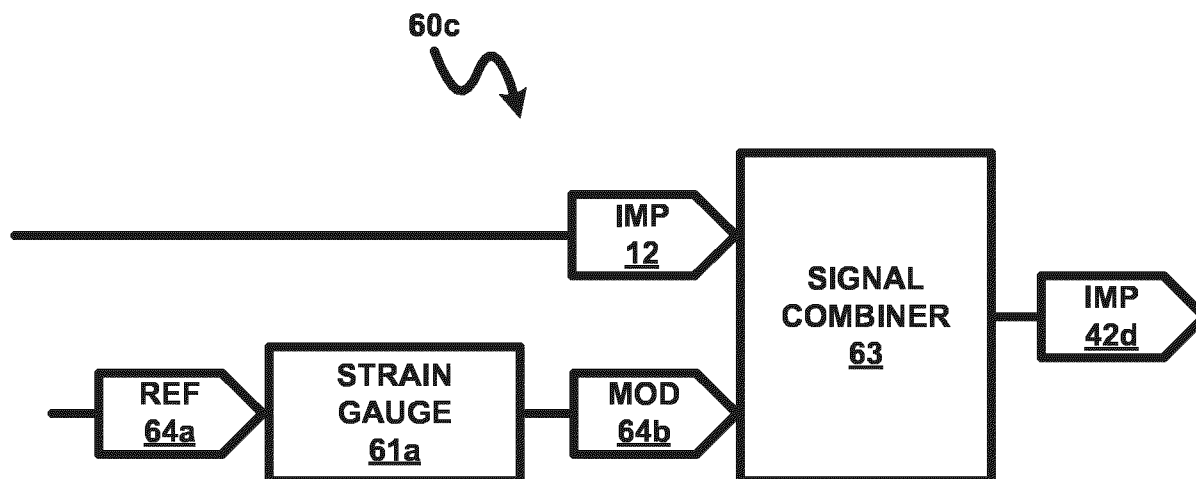


FIG. 6C

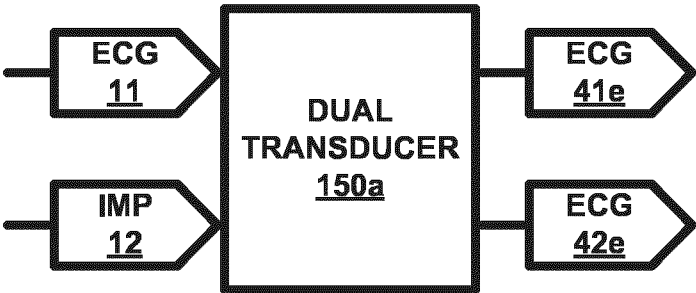


FIG. 7A

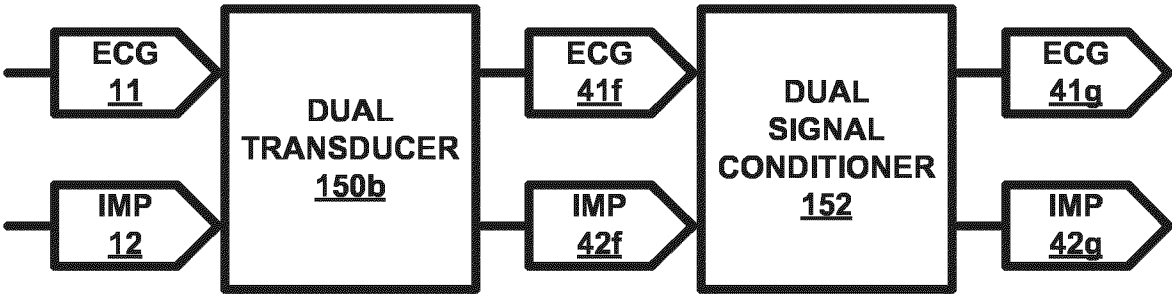


FIG. 7B

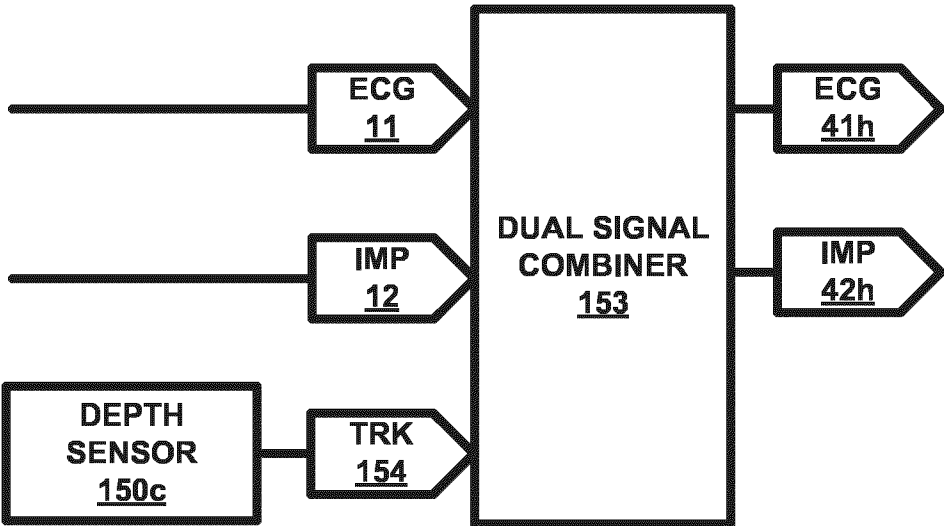


FIG. 7C

## INTERNATIONAL SEARCH REPORT

International application No  
PCT/EP2025/080409**A. CLASSIFICATION OF SUBJECT MATTER**

INV. G09B23/28

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

**B. FIELDS SEARCHED**

Minimum documentation searched (classification system followed by classification symbols)

G09B G09D

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal

**C. DOCUMENTS CONSIDERED TO BE RELEVANT**

| Category* | Citation of document, with indication, where appropriate, of the relevant passages                        | Relevant to claim No. |
|-----------|-----------------------------------------------------------------------------------------------------------|-----------------------|
| X         | US 2023/255537 A1 (TO LOK YIU [US] ET AL)<br>17 August 2023 (2023-08-17)                                  | 1-3,6-11              |
| A         | paragraphs [0005] - [0006], [0029],<br>[0032], [0037] - [0041], [0049],<br>[0052]; figures 2A-2B<br>----- | 4,5                   |
| A         | US 9 633 577 B2 (ZOLL MEDICAL CORP [US])<br>25 April 2017 (2017-04-25)<br>claim 1; figure 1<br>-----      | 1-11                  |



Further documents are listed in the continuation of Box C.



See patent family annex.

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Date of the actual completion of the international search

2 January 2026

Date of mailing of the international search report

15/01/2026

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Authorized officer

Lavin Liermo, Jesus

# INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2025/080409

| Patent document<br>cited in search report | Publication<br>date | Patent family<br>member(s) | Publication<br>date |
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|                                           |                     | WO 2022008301 A1           | 13-01-2022          |
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| US 9633577 B2                             | 25-04-2017          | NONE                       |                     |
| -----                                     |                     |                            |                     |